

course_notebook

June 8, 2026

1 Welcome to the Python course!

This document is called a Jupyter notebook. It consists of sections of code and text called “cells” that may be executed independently.

I’m using this as a scratchpad for testing code live during the course.

```
[1]: import numpy as np
```

1.0.1 Lists

One of the most useful components of python are the **containers**. Containers are used to store objects.

Lists are the most commonly used type of container. They are mutable, ordered, and can contain objects of any type.

<https://docs.python.org/3/tutorial/introduction.html#lists>

```
[2]: # Create a list where the first element is an int, the second is a string, and  
# the third is a float.  
my_list = [0, 'cat', 3.14159]  
  
print(my_list)
```

```
[0, 'cat', 3.14159]
```

```
[3]: # Lists may be changed after they are created.  
# For example the append method adds a new element to the end of the list.  
my_list.append('dog')  
  
print(my_list)
```

```
[0, 'cat', 3.14159, 'dog']
```

1.0.2 Tuples

Another type of container is a **tuple**. Tuples are different from lists in that they are **immutable**, meaning they cannot be changed after they are created.

<https://docs.python.org/3/tutorial/datastructures.html#tuples-and-sequences>

```
[4]: my_tuple = ('a', 'b', 'c')
     print(my_tuple)
```

```
('a', 'b', 'c')
```

```
[5]: my_tuple[0] = 'z'
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[5], line 1
----> 1 my_tuple[0] = 'z'

TypeError: 'tuple' object does not support item assignment
```

Another commonly used container in python is the **dictionary**. A dictionary represents a set of key value pairs.

<https://docs.python.org/3/tutorial/datastructures.html#dictionaries>

```
[6]: my_dict = {'a': 7, 'b': 12, 'c': [1, 2, 3]}
     print(my_dict)
```

```
{'a': 7, 'b': 12, 'c': [1, 2, 3]}
```

```
[7]: # Retrieve the value associated with the key "a":
     print(my_dict['a'])
```

7

1.0.3 Importing Modules and Packages

You can use bring other python files or external python packages into your program using the **import** statement. A few common packages are shown in the next few sections.

1.0.4 Numpy

Numpy is a commonly used scientific computing package for python. It is very useful for things like array or matrix manipulation, statistics, or linear algebra.

```
[8]: import numpy as np

array = np.array([1.0, 5.0, 8.0, 1.234, 101.001])
print(array)
```

```
[ 1.      5.      8.      1.234 101.001]
```

```
[9]: # Compute the mean and standard deviation of this array
print( np.mean(array) )
print( np.std(array) )
```

```
23.247
38.963084354296186
```

```
[10]: # define a 3x3 array
a = np.array([[1, 0, 0],
              [0, 0, 1],
              [0, 1, 0]])

print(a.shape)

# Compute the eigenvalues and eigenvectors of the array
np.linalg.eig(a)
```

```
(3, 3)
```

```
[10]: EigResult(eigenvalues=array([ 1., -1.,  1.]), eigenvectors=array([[ 0.
, 0.
, 1.
],
[ 0.70710678,  0.70710678,  0.
],
[ 0.70710678, -0.70710678,  0.
]]))
```

1.0.5 Pandas

[Pandas](#) is a data analysis package for python. Pandas provides a data type called a “DataFrame”, which can be thought of as similar to an excel table. A DataFrame has an index, columns, and rows.

```
[11]: import pandas as pd

df = pd.DataFrame({
    'A': [1,2,3,4],
    'B': [5,6,7,8],
    'C': [9,10,11,12]
})

df
```

```
[11]:    A  B   C
      0  1  5   9
      1  2  6  10
      2  3  7  11
      3  4  8  12
```

```
[12]: # Add a new a new column by performing a transformation on another column
df['C Transformed'] = df['C'].transform(lambda x: x**x)

df
```

```
[12]:    A  B   C  C Transformed
      0  1  5   9          387420489
      1  2  6  10       100000000000
      2  3  7  11       285311670611
      3  4  8  12       8916100448256
```